

1(a)	any two from: 1 replace chloroplast (suspension) with, water / boiled chloroplast (suspension) ; 2 replace DCPIP with, water ; 3 keep (chloroplast suspension with DCPIP) in the dark ;	2																		
1(b)(i)	<table border="1"> <thead> <tr> <th>percentage concentration of chloroplast suspension</th><th>time taken for DCPIP to decolourise / s</th><th>rate of light-dependent stage of photosynthesis / s⁻¹</th></tr> </thead> <tbody> <tr> <td>10</td><td>351</td><td>2.8</td></tr> <tr> <td>20</td><td>59</td><td>16.9</td></tr> <tr> <td>30</td><td>21</td><td>47.6</td></tr> <tr> <td>40</td><td>10</td><td>100.0</td></tr> <tr> <td>50</td><td>5</td><td>200.0</td></tr> </tbody> </table> rates calculated correctly ; (correct) rates correctly rounded to one decimal place ;	percentage concentration of chloroplast suspension	time taken for DCPIP to decolourise / s	rate of light-dependent stage of photosynthesis / s ⁻¹	10	351	2.8	20	59	16.9	30	21	47.6	40	10	100.0	50	5	200.0	2
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1(b)(ii)	1 axes correctly orientated with labels and rate unit and appropriate scales ; 2 points plotted accurately ($\pm \frac{1}{2}$ small square) ; 3 suitable line drawn ;	3																		

Question	Answer	Marks
1(c)	any two from: (Did work in 10% sucrose solution because) (sucrose solution) has, same / similar, <u>water potential</u> as in the chloroplast ; - water will not enter (chloroplast), by <u>osmosis</u> / as there is no water potential gradient / ora ; <u>chloroplasts</u> will not burst / ora ;	2
1(d)	- one, stated, concentration of chloroplast suspension ; any five from: - minimum of five, stated, temperatures within the range 10 to 60 °C ; - method of maintaining a constant temperature ; - allowing chloroplast suspensions to reach set temperatures (before exposing to light) ; - swirl chloroplast suspension (before exposing to light) ; - method for ensuring same light intensity ; - carry out experiment in dark room ; - colour of sample to match chloroplast suspension (alone) / green colour standard ; - correct <i>ref. to repeats and mean</i> ;	6